

DENTON, TX, USA

WMO: 722589

Lat: 33.206N

Lon: 97.199W

Elev: 642

StdP: 14.36

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 03991

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.5	24.7	8.8	8.8	28.1	12.7	10.7	34.0	26.9	53.6	24.9	53.0	7.7	350	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.8	101.7	74.4	99.4	75.0	97.2	75.0	78.7	92.8	77.9	92.0	77.1	90.9	10.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	134.9	84.0	74.0	130.0	82.9	73.2	126.3	82.3	42.7	92.9	41.6	92.4	40.8	90.8	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.5	21.3	19.4	13.2	104.4	4.4	3.3	10.1	106.8	7.6	108.7	5.1	110.6	1.9	113.0	
			DB												
			WB	11.9	80.2	3.9	1.2	9.1	81.1	6.8	81.8	4.6	82.4	1.7	83.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	44.6	48.7	56.9	64.3	72.9	81.1	84.9	85.1	77.5	66.2	55.6	46.1
	DBStd	16.46	9.85	10.89	9.65	7.86	6.85	4.48	4.15	4.63	6.36	8.68	9.56	9.72
	HDD50	666	224	144	44	4	0	0	0	0	0	4	52	194
	HDD65	2492	634	464	283	106	18	0	0	0	3	91	303	590
	CDD50	6293	57	106	257	434	711	933	1081	1088	825	508	221	72
	CDD65	2644	2	7	31	85	263	483	616	623	378	130	22	4
	CDH74	31585	21	62	239	716	2553	5666	8250	8360	4223	1296	175	24
	CDH80	16092	1	8	40	160	950	2840	4725	4849	2045	440	31	3
Wind	WSAvg	9.2	9.2	10.2	10.7	11.0	10.2	9.5	8.2	7.5	7.5	8.4	9.0	8.9
Precipitation	PrecAvg	36.1	2.2	2.4	3.3	3.9	4.7	3.5	2.2	2.0	2.6	3.7	3.0	2.5
	PrecMax	55.5	5.7	7.4	7.0	9.0	12.8	11.7	9.3	5.0	9.9	10.6	9.9	8.9
	PrecMin	16.7	0.1	0.2	0.0	0.5	0.5	0.5	0.0	0.0	0.0	0.0	0.2	0.3
	PrecStd	9.8	1.4	1.9	1.9	1.9	2.6	2.4	2.0	1.5	2.1	2.9	2.5	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.9	80.8	84.2	88.9	95.3	99.6	104.4	105.9	100.7	92.4	83.7	77.2
		MCWB	58.3	61.5	64.9	70.5	75.1	75.7	74.8	74.4	73.0	71.4	65.2	62.4
	2%	DB	71.9	74.9	80.1	83.9	91.3	97.1	101.5	102.2	96.6	88.8	79.0	71.9
		MCWB	58.1	61.0	64.5	68.1	73.5	75.4	74.7	74.2	73.1	70.7	64.0	59.2
	5%	DB	66.6	71.2	76.3	81.2	88.4	94.5	99.2	99.9	93.2	84.6	74.8	67.6
		MCWB	55.8	59.4	62.7	67.4	72.6	74.8	75.4	74.8	72.9	68.4	63.0	58.6
	10%	DB	62.9	66.6	72.5	78.5	85.3	92.2	96.8	97.5	90.6	81.5	71.5	63.2
		MCWB	53.5	57.6	61.2	65.8	71.1	74.7	75.2	75.1	72.5	67.3	61.4	55.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.3	67.3	69.5	74.3	78.4	80.7	80.0	79.1	78.1	75.7	70.4	66.8
		MCDB	69.8	74.1	78.9	82.2	91.1	94.6	95.1	93.9	90.8	84.6	78.3	72.4
	2%	WB	61.8	65.1	67.4	71.7	76.3	78.4	78.7	78.0	76.6	73.7	67.5	63.9
		MCDB	67.3	71.0	75.7	80.0	86.5	91.5	93.2	93.5	89.0	83.7	73.7	69.5
	5%	WB	58.3	62.4	65.5	70.0	74.9	77.3	77.8	77.2	75.5	72.1	65.6	60.1
		MCDB	64.7	68.0	73.1	77.5	84.2	89.6	92.4	92.8	87.1	81.1	72.0	64.5
	10%	WB	53.6	59.2	63.7	68.3	73.4	76.2	77.0	76.4	74.5	70.2	63.2	55.8
		MCDB	61.3	66.0	70.5	75.4	82.7	87.9	91.1	91.3	85.6	78.0	70.0	62.3
Mean Daily Temperature Range	5% DB	MDBR	22.8	21.9	21.7	22.0	20.5	20.0	21.4	21.8	22.0	23.1	22.3	21.4
		MCDBR	30.5	27.4	25.2	24.3	22.8	22.5	24.4	24.8	24.3	25.8	25.4	26.8
		MCWBR	18.8	16.7	13.5	13.0	9.5	7.2	6.7	6.7	8.0	11.4	14.2	16.5
	5% WB	MCDBR	25.3	23.3	20.5	19.2	19.3	19.5	21.1	21.4	19.9	20.0	21.1	20.5
		MCWBR	18.5	16.7	13.0	12.4	9.7	7.6	7.0	6.6	7.3	10.0	15.1	16.3
Clear-Sky Solar Irradiance	taub	0.312	0.324	0.356	0.389	0.425	0.437	0.438	0.438	0.397	0.358	0.333	0.317	
	taud	2.496	2.472	2.400	2.333	2.296	2.296	2.296	2.329	2.313	2.374	2.470	2.479	2.482
	Ebn at Noon	288	295	292	285	275	270	269	267	275	279	277	279	
	Edh at Noon	27	31	36	40	42	42	40	40	36	31	28	26	
All-Sky Solar Radiation	RadAvg	912	1081	1420	1725	1861	2095	2130	1955	1634	1278	970	798	
	RadStd	92	167	106	139	185	203	142	121	148	190	125	99	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page